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1)

This project really helped me understand more about the capabilities of GenAI. I appreciated the Socratic Guidance method. How a GenAI system can guide learning without direct answers and through making me think logically is a learning method that can greatly benefit me. For example, in almost every one of the responses GenAI would give me an analogy to help me understand, such as “imagine you're trying to find the lowest point of a valley by walking downhill. If you take tiny steps, you gradually reach the bottom. What happens if you take one enormous leap? Where might you land relative to the bottom?” This helped me understand greater why parts of the code did what they did.

It was also able to assess the depth of my responses. If my original reflections did not show much understanding of the topic, it would respond in a way that would explain the topic. This allowed me to not feel worried about how much I knew about specific modules in my model but allowed me to understand it in the end.

GenAI also did an excellent job at identifying which parts of the code I would be referring to. For example, when I gave it an error message “ValueError: target size [480,1] vs input size [120,1],” it knew exactly what part of my code I switched and was able to guide me to understanding it.

What is impressive to me is how GenAI can create new AI for you. It knows how to make code to create data and execute a whole new predictive AI model and can pinpoint exact variable values to use. Just like in the part of my code where it says `random_state=42`, the number seems randomly generated but I know it is there for a specific reason.

2)

I learned many things about building an AI system from this project. Firstly, I better understood the basic data pipeline. I learned how to better generalize where the dataset was getting generated so I could upload it to my git repository for grading. Another interesting thing was how there is a ton of code you can change, and the AI will still run, but only through testing can you figure out that it is not as accurate or working as you intended. Finding a sweet spot for training is important too. You want to have enough data for training and the model to find patterns in your data, but also make sure the data is correctly correlated. There are many variables such as the optimizer which are extremely important to maintain accuracy in the model and there are different optimizers available.

It felt very satisfying to actually understand what was happening in each step of my model throughout the debugging process in the project. I learned the importance of all the modules, the optimizer, PyTorch, ReLU, more about the Sigmoid function, BCE for binary classification, and learning rate sensitivity.

Perhaps the most interesting realization is how hard it may be to detect silent bugs. It makes me wonder about large scale AI models, and how hard it may be to detect these bugs that do not produce error messages.

Overall, this project helped me deepen my understanding of GenAI and creating an AI system in general. It allowed me to apply what we have learned in class and feel satisfied with the time I spent creating it all.